

22. Use case: Player Queues against Random player

Iteration: 1

Primary Actor: User

Goal: Allow the player to queue for a game and find an opponent.

Preconditions:

- Player is logged in and has selected a game.

Trigger: Player clicks “ Queue against Random Player”

Scenario:

1. Player selects a game from the library.
2. Player clicks “Queue for Match”
3. System adds the player to the matchmaking queue.
4. System searches for opponents with similar skill levels.
5. System notifies the player when a match is found.

Exceptions:

1. No opponents are available within the skill range.
2. Player disconnects while queued.

Priority: High

When available: Within 3 sprints

Frequency of use: Frequently (every time a player queues for a game).

Channel to actor: GUI (game panel UI)

Secondary actors: N/A

Channels to secondary actors: N/A

Open issues:

1. How will the system handle players who disconnect while queued?

23. Use case: Find opponent based on ELO

Iteration: 1

Primary Actor: Matchmaking System

Goal: Match a player with an opponent of similar skill level using ELO rating.

Preconditions:

- A player has queued for a game.
- At least one other player is also queued in the same game type.

Trigger: A player enters the matchmaking queue.

Scenario:

1. The system checks the queue for an opponent with X ELO range.
2. If an opponent is found within the range, the match is created.
3. The system places both players in the game lobby.
4. if no opponent is found, the system waits for Y seconds before deciding whether to expand the search range.

Exceptions:

1. Matchmaking server is down.
2. Player cancels the queue before a match is found.

Priority: High

When available: Sprint 1

Frequency of use: Frequently (every time a players queues)

Channel to actor: System process

• The system checks for an opponent within the ELO Range, but it doesn't specify whether the range dynamically adjusts based on wait time!

↳ If a match is not found, should the search expand over time?

Secondary actors: N/A

Channels to secondary actors: N/A

Open issues:

1. Should players be notified when the system expands the range?

24. Use case: Expand Search Range

Iteration: 1

Primary Actor: Matchmaking System

Goal: Ensure players find matches by gradually expanding ELO requirements

Preconditions:

- A player has been in the queue for longer than Y seconds without finding a match
- No opponent was found within the initial ELO range.

Trigger: "Find Opponent based on ELO" fails to find a match.

Scenario:

1. The system checks if the player has been waiting for more than certain seconds.
2. The system expands the allowed ELO range(e.g., +50 -> +100 ->200).
3. The system searches again for an opponent within the new range.
4. If a match is found, both players are placed in the game lobby.
5. If still no match is found, the system continues expanding the range in intervals.

Exceptions:

1. If all available opponents disconnect, the system stops searching.

Priority: High

When available: Within 2 sprints

- Instead of just asking if players should be notified, consider adding a UI indicator that warns players when system is broadening search range.

29

Frequency of use: Occasionally (only when matchmaking takes too long)

Channel to actor: System Process

Secondary actors: N/A

Channels to secondary actors: N/A

Open issues:

1. Should the player be notified when the search range expands?

35. Use case: Leaderboard Updates After a Game

Iteration: 1

41

Primary Actor: Game Logic System

Goal: Update player rankings and leaderboard after a game ends.

Preconditions:

- Game has ended, and the outcome (winner, loser) is determined.

Trigger: Game Logic System sends the game outcome to the LeaderboardSystem

Scenario:

1. The Game Logic System sends the game outcome (winner, loser) to the LeaderboardSystem.
2. LeaderboardSystem updates the rank of the winner and loser.
3. LeaderboardSystem updates the leaderboard display

Exceptions:

1. Game outcome data is corrupted or missing.
2. LeaderboardSystem fails to update rankings.

Priority: High

When available: Within 4 sprints

Frequency of use: After every game

Channel to actor: GUI (game panel UI)

Secondary actors: None

Channels to secondary actors: None

Open issues:

1. How will the system handle ties or draws in a game?
2. How will the system notify players when their rank changes?

Iteration: 1

Primary Actor: Player

Goal in Context: Allow the player to view the leaderboard

Preconditions: Player is logged in.

Trigger: Player clicks "View Leaderboard."

Scenario:

1. Player clicks "View Leaderboard"
2. System retrieves the leaderboard data.
3. System displayed the leaderboard to the player.

Post conditions: Player views the leaderboard

Exceptions: Leaderboard data is unavailable

Priority: Medium

When Available: Iteration 1

Frequency of Use: Frequently (when players want to check rankings.

Channel to Actor: In-game interface

Secondary Actors: None

Channel to Secondary Actors: None

Open Issues:

1. How many players should be displayed in the leaderboard at once?
2. How will the system handle leaderboard data if the database is temporarily unavailable?

3. Are players able to report or flag suspicious rankings? (e.g. 250 wins for 250 games played)

Channel to actor: Game user interface

Secondary actors: N/A

Channels to secondary actors: N/A

Open issues:

1. How detailed should the statistics collected be?
2. How many match histories should be stored at a time?

39. Use case: Filter Leaderboard by Wins or ELO

Iteration: 1

Primary Actor: Player

Goal: Allow the player to filter the leaderboard depending on ELO or Wins

Preconditions:

1. Player is logged in
2. Leaderboard database is operational

Trigger: Player presses on drop down menu and selects whether to filter to leaderboard by wins or elo

Scenario:

1. Player navigates to the leaderboard page
2. Player clicks on the drop down menu
3. Player selects one of the two options being: "Filter by Wins" or "Filter by ELO"
4. Depending on the request the system fetches the appropriate data from the database

46

5. The system displays the leaderboard based on the player's preference
6. Player view's the filtered leaderboard

Exceptions:

1. If the database is unresponsive or takes too long to respond and error message will be displayed

Priority: Medium

→ *this feature is classified as medium priority. But filtering leaderboard data needs to be high priority as it might be critical if players want quick access to rankings*

When available: 1st or 2nd Iteration

Frequency of use: Occasionally, when players want to track where they stand against the competition

Channel to actor: Game user interface

Secondary actors: N/A

Channels to secondary actors: N/A

Open issues:

1. How should we approach the process of fetching the data in order to improve speed and performance?